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Multi-Agent Systems for Manufacturing Digital Twins: A Perspective on Agency and Large Language Models

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Abstract: The capability of Large Language Models (LLMs) and LLM agents to understand and generate natural language represents a major advance in the evolution of software agents, allowing not only opportunities for human interaction with manufacturing systems and their digital twins, but also language-dependent tasks ranging from predictive modeling to workflow analysis. This paper contrasts the capabilities of classic autonomous software agents and LLM software agents and offers a multi-agent framework for harmonizing autonomous software agents and LLM agents within a digital twin-enabled hybrid manufacturing system for the repair and restoration of damaged parts with complex forms. These multi-agent systems represent a useful paradigm to control process-driven systems, allowing both dynamic control of (re)manufacturing at a process level and human oversight and interaction at the system level.

Keywords: Digital Twin, Manufacturing System, Large Language Models, Multi-Agent Systems

1. INTRODUCTION

Interest in agent-based systems and digital twins has surged with the rapid development of Generative AI and Large Language Models (LLMs) to generate content that accurately represents complex patterns of natural language, sounds and imagery. Powered by deep learning architectures such as transformers and trained on large datasets, LLMs and other Gen AI models can perform a wide range of tasks such as answering queries, summarizing texts, and generating creative content. Their natural language facility coupled with new capabilities for agentic control enables direct interactions between human users and LLMs—a major benefit for incorporating human oversight and control of the dynamic behavior of large and complex digital-twin enabled manufacturing systems.

Recent excitement about LLM agents follows on the heels of an earlier generation of research in autonomous software agents for industrial applications. Both types of agents are capable of making decisions and performing specific tasks to achieve goals without human supervision. But differences in architecture, design philosophy, and functionality distinguish classic autonomous software agents from current LLM agents. Autonomous software agents are best suited for complex problem-solving and control in decentralized systems with large numbers of interacting entities and where numerical data is the currency of communication. In contrast, LLM agents are designed to generate natural language and, accordingly, can infer intent and converse intelligently with humans.

While LLM ability to understand and generate language is a major advance, they are not designed for complex math calculations requiring high precision and accuracy, and struggle with problems requiring a deep understanding of mathematical relationship and abstract concepts typical of industrial applications. Similarly, because of their reliance on predefined algorithms and extremely large and generalized datasets with significant training requirements, current LLMs are not viewed as well suited for tasks requiring near real-time response to system changes on order of minutes, even seconds.

This paper offers and evaluates a multi-agent framework that harmonizes both intelligent software agents and LLM agents in a digital-twin enabled manufacturing system for the repair of high-precision parts such as turbine blades Sensor-based communications and interactions between and among physical processes and their digital twins are enabled by autonomous intelligent software agents, while LLM agents support system objectives through system monitoring, incorporation of relevant external data, and interaction with the human user for directed actions. Contributions of this paper include:

1) Muti-Agent System comprised of autonomous software agents and LLM agent platforms for coordination and control of a multi-process (re)manufacturing system.
2) Mechanisms for harmonizing text and quantitative process data for messaging between the digital twin system and LLM agents to facilitate human oversight and input.
3) Case study application to repair and restore complex shaped damaged parts using multi-process hybrid manufacturing (c.f. gearboxes, turbine blades and shafts).
4) Assessment of the narrowing capability gap between LLM agents and autonomous software agents with respect to their ability to manage physics-based and data-driven model inputs and outputs, as well as recent research expanding opportunities for harmonized agent control.

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The remainer of the paper is organized as follows. Section 2 describes research related to advances in agent technology from the perspective of manufacturing system control. Section 3 motivates the problem. Section 4 introduces a multi-agent framework that harmonizes autonomous multi-agent control with humans-in-the-loop. Section 5 describes the case study application followed by a discussion of domain-knowledge integration in Section 6. New research suggesting larger roles for LLMs in physics-based and data-driven applications is introduced in Section 7. Conclusions are offered in Section 8.

2. RELATED WORKS

The concept of agency within the frame of computer science, and by extension the world of software agents, continues to evolve. Software agents are computer programs that act on behalf of a user *in a relationship of agency* by engaging in reactive and proactive behaviors *on behalf of that user*.

Intelligent Software Agents. Early software agents were simple programs, usually for information retrieval. The first practical implementations of intelligent software agents were ruled-based systems developed in the 1960s and referred to as expert systems. Early autonomous software agents, like those today, were conversant in numerical data such as sensor readings, predictive outputs of physical models and quantitative parameter values—and typically relied on predefined rules, algorithms and machine learning models for task management.

Today's autonomous software agents can assume a range of capabilities including: 1) persistence in that the program is not executed on demand but runs continuously and decides for itself when to perform a task or activity; 2) autonomy in that the agent can make decisions about task selection, prioritization, goal-directed behavior and decision-making, all without human intervention; 3) social ability in that the agent can engage with other entities through some sort of communication interface; and 4) reactivity in that the agent can perceive its environment and to react to it appropriately.

A multi-agent system or MAS, as its name suggests, is comprised of multiple autonomous software agents that are capable of interacting together and solving problems collaboratively. The concept of a system of agents was formally introduced in the 1980s concurrently with the concept of distributed artificial intelligence (DAI). DAI is a system where multiple independent software agents work across multiple distributed locations (or computer nodes) to achieve a common goal, distributing the computational load and decision-making authority. DAI emerged as an approach to solve large complex scientific and engineering problems at scale and asynchronously (Wooldridge, 1999).

LLMs and LLM Agents. LLMs such as GPT-4o, LLaMA 3.4, and Claude 3.5 Sonnet represent a class of deep learning architectures called transformer networks (Zhao et al., 2023). A transformer is a type of neural network that learns language and context by tracking relationships in sequential text data. LLMs are autoregressive in that they predict the next word in a sequence of words based on the words that came before it. Trained using unsupervised machine learning on very large volumes of non-proprietary text found on the web and in public databases, these models offer the ability to generate and analyze text in new ways.

Since LLMs are trained on a single static, centralized and generalized datasets they are able to learn complex language patterns and "understand" context. LLMs and LLM agents are able to construct language as if it were produced by a human and engage in a conversational style of interaction. And unlike intelligent software agents that perform specific tasks based in narrow knowledge sets, LLM agents embody cross-domain knowledge that reflects the breadth of human intelligence.

LLM agents have generated excitement because they are able to generate a trace of their "reasoning" during problem-solving, referred to as chain-of-thought reasoning, which provides a trail of how an output or conclusion was reached (Wei et al., 2022). Chain-of-thought prompting of the LLM emulates human-like reasoning and offers a structured mechanism for problem solving, planning, and other complex industrial tasks such as machine fault diagnosis, quality assessment and workflow generation by enabling a single prompt or task to be broken down into smaller, manageable subtasks. A comparison of classic intelligent software agents and LLM agents is provided in Figure 1.

DIMENSION OF AGENTNESS	CLASSIC INTELLIGENT SOFTWARE AGENTS	LLM AGENTS
Core Functionality	Perform specific tasks with minimal human intervention	Generate and process natural language text (also audio and graphics)
Decision-Making	Capable of real-time, independent decision-making	Decision-making is guided by input prompts, usually human, and is context-based.
Adaptability	Quickly adapts to dynamic environments and learns through experience	Limited. Adapts based on RAG or fine-tuning with additional data
Learning Mechanism	Machine learning (reinforcement learning), rule-based AI	Pre-trained on large datasets, fine-tuning for specifics
Interaction with Users	Limited to none, typically task-specific program adjustments	Highly interactive, focused on conversational AI with prompts
General Applications	Scientific applications such as robotics, automation of equipment, manufacturing processes, etc.	Content generation, software coding, chatbots and customer support
Key Limitation for Practical Use in Manufacturing Systems	Limited natural language capabilities	Limited numerical data capabilities. Lack of real-world autonomy and decision-making ability

Figure 1. Comparison of autonomous agents and LLM agents

3. PROBLEM DESCRIPTION

The introduction of LLMs and LLM agents launched a dialogue among researchers about their compatibility and synergy with existing software technologies. This paper represents one facet of that conversation. In particular, multi-agent systems have a long history in industrial applications. Research in autonomous multi-agent systems spiked from 1990 to 2010 and continues today for applications such as smart manufacturing (Kim et al., 2020), scheduling (Zhang et al., 2022) and management of production flows (Komesker et al., 2022). An early discussion of the state-of-the-art of agent-based manufacturing systems is provided by Leitao (2009). Pulikottil et al. (2020) provide a review of current trends.

Recent renewed focus on digital twin systems has raised interest in LLM synergies with MAS—where a digital twin system includes not only the physical system but also its high-fidelity digital representation, as well as bi-directional feedback and control mechanisms between the two. This structure leads to a large number of interactions (i.e. physical-physical, virtual-virtual, and virtual-physical) and data types (e.g. CAD data, sensor data, process data, etc.) which create critical challenges for interoperability, synchronization and closed-loop feedback. Pretel et al. (2024) explored the synergies between MAS and digital twins in a recent literature review. Ocker et al. (2021) and Vogel-Hauser et al. (2021) embed digital twins within a MAS using the Asset Administration Shell platform developed by the German Industry 4.0. Related to the application here, Latsou et al. (2023) and Zheng et al. (2020) offer approaches to integrate MAS with digital twins for anomaly detection and manufacturing, respectively. Figure 2 provides a stylized schematic of an agent-enabled digital twin system.

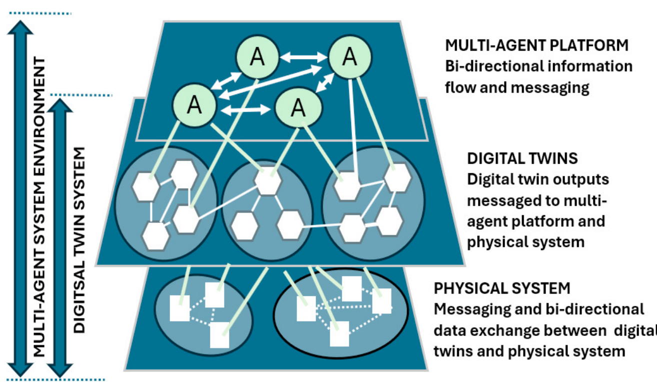

Figure 2. Stylized schematic of a MAS and its many interconnections

LLMs and LLM agents represent a further advance in systems control. While research remains nascent, initial results suggest that LLMs and LLM agents, when embedded with autonomous agents can assume contributing roles in the coordination, control and optimization of digital twin-enabled manufacturing systems (Huang et al., 2024; Lim et al., 2024; Han, 2024). Specific frameworks for industrial applications have been proposed. Gao et al. (2024) explored models that bridge the gap between human operators and mechanical processes in determining tool wear. Fu et al. (2024) combine LLMs with a vision-language model for detecting slab defects during the smelting process. Most recently, and related to the framework proposed here, Xia et al. (2023) and Xia et al. (2024b) leverage LLMs and LLM agents in the planning and control of production systems and automation systems, respectively. These applications can be modularized and have repeatable operations that are easily routinized.

This paper proposes a framework that harmonizes LLMs and LLM agents with a digital twin system for process-driven (re)manufacturing where each part/damage is unique. The digital twin system is composed of physics-based and data-driven models for maintaining process stability and LLMs that provide generalized information about material properties and behaviors. Queries to the digital twin system allow the human operator to combine domain-specific knowledge provided by the LLM with process performance indicators to identify and evaluate corrective actions. Of particular note, the ability to manage the interface of text, localized in the LLM, and numerical/ordinal data, the province of the physical system and digital twin, remains to be fully resolved. In the proposed framework, numerical information from the physical process and its digital twin is converted to natural language for understanding by the LLM. While this approach can satisfy applications where numerical data is understood as a "number word" it does not fully satisfy applications such as process control where an ordinal sense of the "number word" can be critical to a control decision, as will be discussed in Section 6.

4. METHODOLOGY

Figure 3 provides a systems view of the MAS framework comprised of a *Digital Twin System* and an *LLM Agent System*. The *Digital Twin System* is composed of the processes/assets that comprise the *Physical Processes* and the sensors and actuation that effect dynamic control of the overall physical system and the *Digital Twins* that model process behavior.

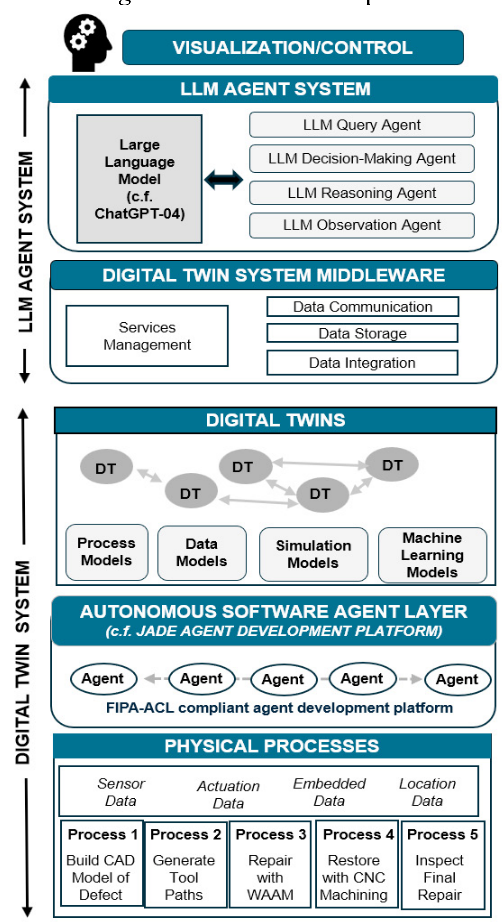

Figure 3. MAS Framework for (Re)Manufacturing

Bi-directional communication and actuation between the physical system and its digital twin is managed by an *Autonomous Software Agent Layer*. Autonomous agents manage the numerical inputs and outputs of the physical processes and their digital twins, continuously analyzing real-time data from the physical system to detect process instabilities, predict anomalies, recommend adjustments and take corrective actions or trigger the necessary responses without direct human intervention to restore stability and optimize performance (c.f. avoiding chatter during machining or uneven layer deposition during AM). JADE (Java Agent Development Program) is a well-known commercial middleware for developing complex autonomous agent systems and remains one of the most popular open source agent development platforms in use currently (and since the early 1980s). JADE implements standardized messaging adhering to the FIPA-ACL standards for interoperability. The *LLM Agent System* is composed of an LLM model, such as GPT-4o, DeepSeek-R1, or LLaMA 3.4, that contributes generalized knowledge to the human operator. The LLM model may be supported by fine-tuning, retrieval augmented generation or prompt engineering to incorporate domain knowledge specific to the processes. The LLM agents facilitate human interaction with the LLM, responding to queries and executing tasks from the human, as well as generating relevant domain-knowledge for human decision-making. LLM agents assume functions that align with the specific requirements of the application. A *Digital Twin System Middleware* manages data communication, integration and storage between the *LLM Agent System* and the *Digital Twin System*.

5. CASE STUDY IN (RE)MANUFACTURING

A hybrid manufacturing system for the repair and restoration of critical metal components with very tight tolerances such as turbine blades for the aerospace and energy sectors serves as the case study. The repair and restoration process follows a step-by-step sequence where the part is passed from one process to another. Each damaged part is unique in material, geometry of damage, and location of defect requiring a synthesis of human intelligence supported by LLM knowledge, as well as the accumulated history of past repairs for similar damage. The physical process workflow is shown in Figure 4. The repair and restoration sequence starts with scanning the damaged component using contact-free measurement techniques such as laser scanning and structured light 3D sensing. Scanning generates a point cloud of the damaged component which can be compared with the undamaged component to extract a 3D model of the defect (1). Then, tool paths for additive and subtractive processes are generated (2).

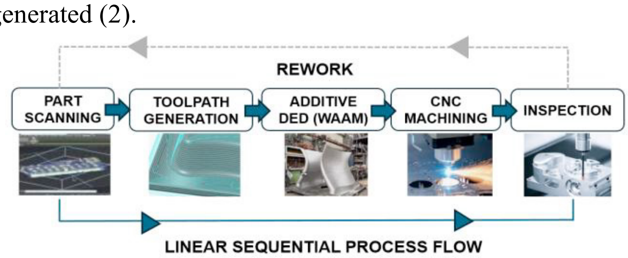

Figure 4. Repair and restoration workflow

The additive and subtractive processes are applied in sequence to repair and restore the damaged part. Wire Arc Additive Manufacturing (WAAM) adds material to fill the damage (3). CNC machining removes excess material to restore the component's original geometry and surface quality (4). Finally, the component is inspected (5). If the part fails to meet specifications, the repair process is repeated. A discussion of the repair and restoration tasks performed by the MAS, as shown previously in Figure 3, is provided below.

Physical System. The *Physical Processes* are automated to maximize seamless transitions across the processes. The additive and subtractive processes are provided by a single hybrid machine which overcomes the imitations of each individual process and capitalizes on their strengths. The hybrid machines are equipped with in-situ monitoring systems (cameras, laser scanners, etc.) to allow continuous inspection of the additive build process and provide real-time data to correct any deviations from the desired geometry.

Digital Twins. The *Digital Twins* are comprised of physics-based and data-driven models that mirror the process and identify any parametric corrections to be implemented by the *Autonomous Software Agent Layer*. For hybrid manufacturing typical parameters include layer thickness and scanning speed and. for machining. cutting speed and feed rate. The goal is to ensure the highest quality parts in terms of microstructure, mechanical properties, tolerances, surface finish and integrity. Digital twins also monitor MAS status and respond to requests from LLM agents for information about system performance.

Autonomous Software Agent Layer. Autonomous agents are a communications bridge between the digital twins and physical processes and implement needed corrective adjustments to the physical system as needed. Tasks might include: 1) managing the coordinate frames during scanning to assure dimensional part quality from scan through final inspection; 2) implementing changes in process parameters such as deposition layer thickness for WAAM or cutting speeds for CNC machining; and 3) managing settings on in-situ vision systems to allow continuous inspection.

LLM Agent System. The (fine-tuned) LLM and LLM agents interact with the human operator by responding to queries and initiating tasks related to system behavior. Agents include query agents that provide updates on physical system status, observation agents that receive and broadcast information about process deviations, and reasoning agents that resolve conflicting information from the MAS and external sources.

6. LLM DOMAIN-SPECIFIC KNOWLEDGE

From a managerial perspective, generalized LLMs are not sufficient for most industrial applications because they lack critical domain knowledge. When specialized domain knowledge is missing in an LLM's knowledge base, or there are gaps in its training data, LLMs extrapolate (c.f. hallucinate) and provide output that is misleading or inaccurate. And, as noted earlier, LLMs are not designed for reasoning with numbers and complex math calculations requiring high precision and accuracy. Both issues require resolution before LLMs can be reliably used in many industrial applications and are a topic of further research for this team. To address the lack of domain knowledge, fine-tuning (Jeong, 2024), prompt engineering (Shin and Ramanathan, 2024) and Retrieval-Augmented Generation or RAG (Li et al., 2024) have been developed, as summarized in Figure 5. These methods can help inject meaningful numerical data into the LLM model.

As noted earlier, in this framework numerical information is converted to natural language for understanding by the LLM and then transformed back. Alternatively, prompt engineering and fine-tuning can be tweaked to inject numerical knowledge into the LLM model. Prompt engineering involves strategically structuring ("crafting") the question-answer query format in each prompt to shape the model's output. Fine-tuning involves training the pre-trained LLM on a smaller specialized dataset, for example a table of physical model outputs. In a non-LLM approach, Zheng et al. (2020) used semantic engineering (ontology) to define a shared vocabulary among agents for both text and numerical data. Depending on the problem, a more direct, but costly, approach is to create one or more domain-specific Small Language Models (SLMs) that can be updated continuously at lower cost. In addition to the research challenges posed, these solutions remain at a distance from real-world use due to cost, reliability issues and, since this work is customized, lack of in-house AI expertise.

7. NEW RESEARCH EXPANDING LLM CAPABILTY

Very recent research suggests that LLMs and LLM agents may be able to perform tasks that have typically been assumed for autonomous software agents because they are conversant in numbers and relationships between them, or done manually.

Multiscale Data Fusion. For a problem in multiscale materials failure, Beuhler (2024) addressed the question whether a single generalized and fine-tuned LLM is better at connecting knowledge across disparate disciplines, scales and modalities than domain-specific SLMs. The MechGPT model outperformed smaller SLMs. In an illustrative (and fanciful) example, MechGPT was asked to create an analogy between the fracture of a ductile material and a musical composition, especially discussing the concept of counterpoint. To which the MechGPT agent intriguingly replied "the analogy between fracture of a ductile material and music, particularly counterpoint, lies in the interaction of many individual elements that contribute to the overall failure process. Just as in music, the sum of the individual defects is greater than the parts, and the material fails when the combined effects exceed its strength."

Prediction and Anomaly Detection. Recent research by Alnegheimish et al. (2024) takes advantage of the autoregressive nature of LLMs to detect process anomalies in univariate time series. Numerical data were converted to LLM-compatible text-based inputs through a series of reversible transformations. The LLMs were then pre-trained strictly with text using GTP 3.5. While the deep learning models outperformed the LLM models, the LLM models compared favorably with classical methods, beating moving average models and slightly underperforming ARIMA models.

Parameter Selection. To automate selection of parameters for simulation models, Xia et al. (2024b) created a system of LLM agents that dynamically interacts with a digital twin simulation model to explore parameterization possibilities and determine feasible parameter settings. Similar to the process by which a human explores options to select the "best" solution, multiple simulations were performed with different parameter settings using an iterative approach to converge on a best solution. Also, in a multi-agent approach, distinct roles were assigned to the LLM agents regarding their interaction with the simulation model. An observation agent collected data from the digital twin which was then analyzed by a reasoning agent. A decision agent generated an actionable decision, and a summarization agent compiled a report.

Workflow Generation. Creation of workflows is a critical task in the design of an MAS, ensuring that LLM-based AI agents operate effectively and reliably to achieve system objectives. Typically, these workflows are drafted manually as part of overall agent system design. Li et al. (2024) developed a framework for automatically generating workflows that outperformed manually designed workflows and performed complex tasks with a high degree of reliability. Similarly, LLMs have been shown to reduce the burden of workflow generation for the machine learning models embedded in the digital twins manufacturing systems (Gu et al., 2024).

8. CONCLUSIONS

This paper offers a digital twin-enabled framework for autonomous control of a process-driven manufacturing system for an unstructured and not easily modularized or routinized application in part repair. LLMs and LLM agents serve as a "front end", allowing humans to monitor, query and interact with the digital twin physics-based and data-driven models using natural language. LLMs support "what if" scenario analyses that provide not only context for recommended actions, but also adjustment of system parameters and an external check on the feasibility and advisability of that adjustment. LLMs and LLM agents may ultimately provide a common language for inter-process communication as well.

Recent research suggests that the breadth of LLM utility extends beyond the conversational interfaces and natural language applications that we immediately think of as suitable applications for LLMs. As described in the paper, very recent research has demonstrated that LLMs and LLM agents are able to perform many of the tasks that are typically assigned to autonomous software agents or other methods—in particular multiscale data fusion, anomaly detection and time series prediction, parameter selection and workflow generation among others.

LLMs and LLM agents still face challenges in real-world manufacturing applications due to lack of domain-specific knowledge, as well as implementation issues with respect to specialized middleware and, especially, the ability to accommodate numerical/ordinal data from physical processes. For complex physical systems, LLM agent collaboration with autonomous software agents will be essential. The direction of recent research and that of this team to address these issues suggests further advances towards a harmonized future for autonomous and LLM agents for manufacturing control.

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